

MATH 180 Strategies of Problem Solving

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Last time

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- we talked about static vs. dynamic algorithms

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- we found good patterns for static algorithms

Break the Cube

Activity (60 mins)

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- play a few games
- **record data:** how many clues until a correct guess

Break the Cube: Debrief

Let's combine the data ...

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Question

How many clues do we need to find a solution?

Break the Cube: Solitaire

Problem

Suppose we have the clues:

- B: green, blank, blank
- C: yellow, gray, blank
- J: green, blank, blank

with gray=small, green=medium, and yellow=large.

Find the block setup.

Build your own

Problem

Create a Break the Cube solitaire problem, meaning:

- specify the colors of the small, medium, large blocks
- enough clues to determine a **unique** setup

Activity

Swap your puzzle with another group.

- find a solution of their puzzle
- explain why the solution is unique (or find another)

Discussion

Good question

What kind of interesting questions can you think of about Break the Cube?

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Question

Suppose we wanted to try to count the number of possible arrangements of the three blocks. How might we begin to go about it?

Question of the Day

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Puzzle

Suppose there is an enemy submarine at an unknown location on the integer number line, and each minute, it moves at an unknown, fixed integer velocity. You may fire one missile per minute somewhere on the number line in an attempt to hit the submarine. Is there a strategy to guarantee you will eventually hit the submarine?

Final thoughts

Reflection

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- What was the static pattern you found that worked the best?
- What dynamic algorithm would you suggest to improve your static algorithm.

Submit your responses by **Tonight!**